Robot Learning: From Fundamentals to Foundation Models

Lecture 3: Imitation Learning

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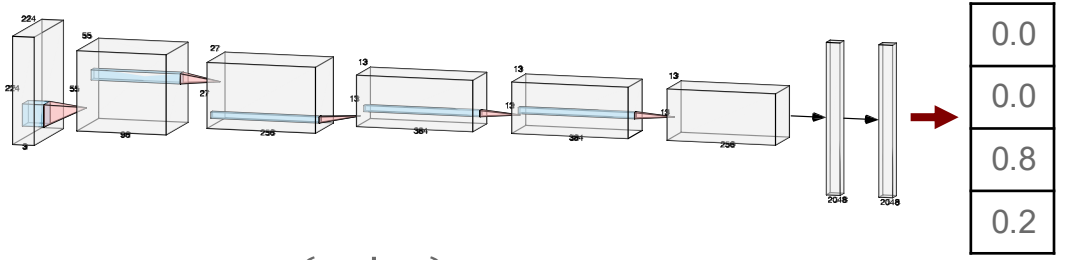
Feedback → Actions

- Moved from 3 papers to 2 papers per lecture
- Extra paper slots on Thursdays in IFW A 32.1
- HW 2 released, due on March 12th
- We can see when you use AI in autograder

Recap: Mapping Observations to Actions



o_t

$\pi_\theta(a_t|o_t)$ **Policy - Partially Observable**

$\pi_\theta(a_t|s_t)$ **Policy - Fully Observable**

a_t

$\boldsymbol{a_t}$ **-** action: the decision taken by the agent at time t

$\boldsymbol{o_t}$ **-** observation: what the agent observes at time t

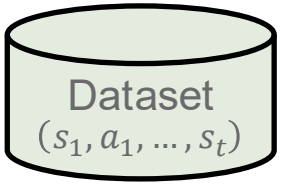
$\boldsymbol{s_t}$ **-** state: the "state" of the world at time t

$\boldsymbol{\tau}$ trajectory: sequence of state and actions $(s_1, a_1, \ldots, s_T, a_T)$

$r(s, a)$ reward: how good is s, a ?

Imitation Learning

- **Data**: Given trajectories collected by an "expert" "demonstrations" $\mathcal{D} = \{(s_1, a_1, \dots, s_t)\}$



- **Goal**: Learn a policy π_θ that "imitates" the expert's behavior

Behavioral Cloning

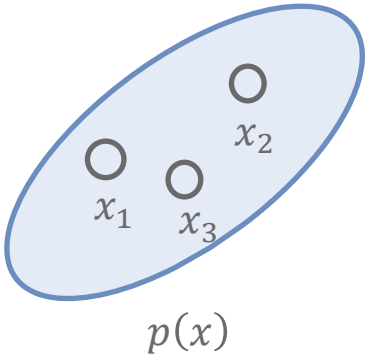
1. Given expert demonstrations: $\mathcal{D} = \{(s_1, a_1, \dots, s_t)\}$

2. For deterministic policy, regress to expert's actions:
$$\min_{\theta} \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{s},\mathbf{a}) \in \mathcal{D}} ||\mathbf{a} - \hat{\mathbf{a}}||^2 \text{ where } \hat{\mathbf{a}} = \pi_\theta(\mathbf{s})$$

3. Deploy policy π_θ on robot

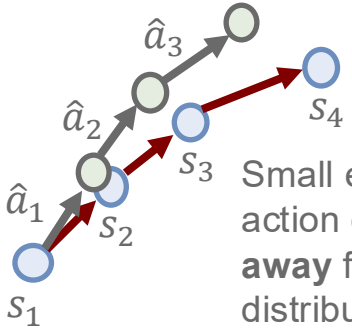
What can go wrong?

Supervised Learning



Inputs are independently, identically distributed (**i.i.d.**) and **independent of predicted labels**!

Behavioral Cloning



Small error in predicted action can lead to **drift away** from training data distribution!

$$p_{expert} \neq p_\pi(s)$$

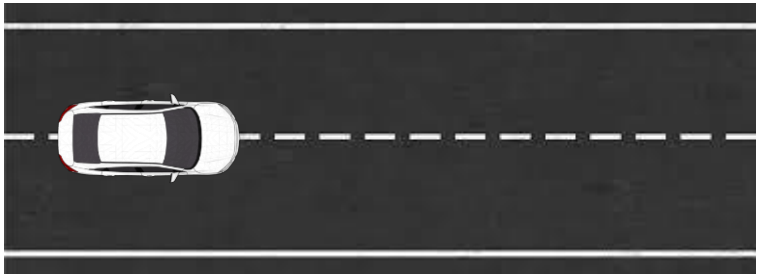
States visited by expert

States visited by the policy

Does it work?

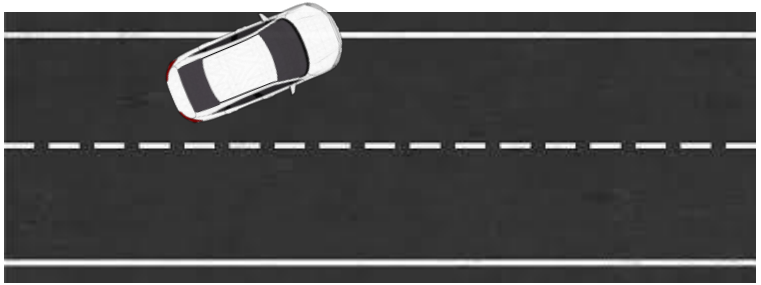
End to End Learning for Self-Driving Cars
Bojarski, et. al., 2016

Why did it work?



Why did it work?

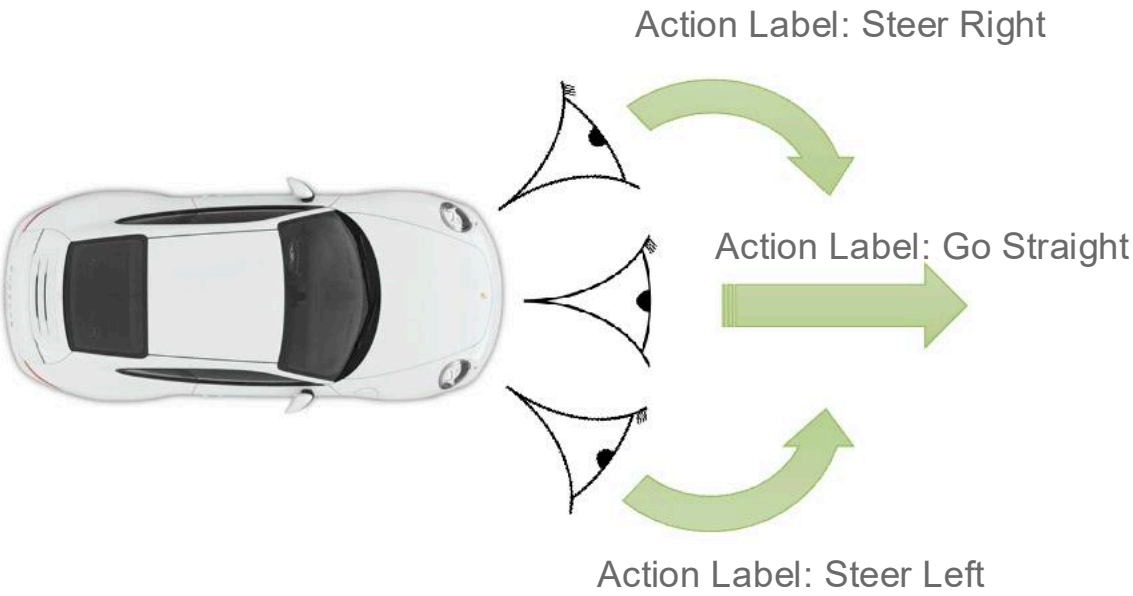
What should the policy do here?
Expert never “visited this state”!



Why did it work? Data Augmentation!

- Add “fake” data that illustrates corrections with side-facing cameras

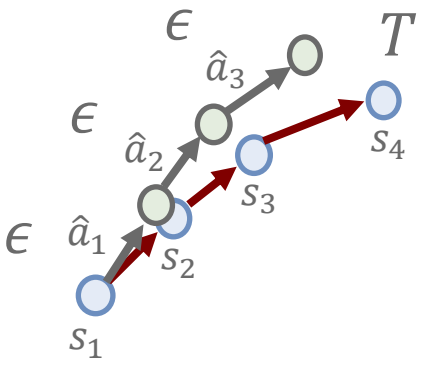
$$\mathcal{D}_{\text{aug}} = \{(\mathbf{o_{center}}, \mathbf{a}), (\mathbf{o}_{left}, \mathbf{a} + \delta), (\mathbf{o}_{right}, \mathbf{a} - \delta)\}$$



Learning to Fly in Swiss Forests

- Same camera trick

Upper Bound of Behavioral Cloning



T: time horizon

ϵ: probability of π_θ making a mistake at any steps

$$c(\mathbf{s}, \mathbf{a}) = \begin{cases} 0 & \text{if } \mathbf{a} = \pi^\star(\mathbf{s}) \\ 1 & \text{otherwise} \end{cases}$$

$$P(\,\mathbf{a} \neq \pi^\star(\mathbf{s}) \mid \mathbf{s}\,) \leq \epsilon, \quad \forall \mathbf{s} \in \mathcal{D}_{\text{train}}$$

$$\mathrm{E}\left[\sum_t c(s_t, a_t)\right] \leq \epsilon T + (1-\epsilon)(\epsilon(T-1) + \cdots) \approx \mathbf{O}\left(\boldsymbol{\epsilon}\mathbf{T}^2\right)$$

Distribution shift causes the error to grow **quadratically!**

More analysis: “A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning” by Ross et al., 2011.

Addressing Compounding Error

Can we make?

$$p_{expert} = p_{\pi}(s)$$

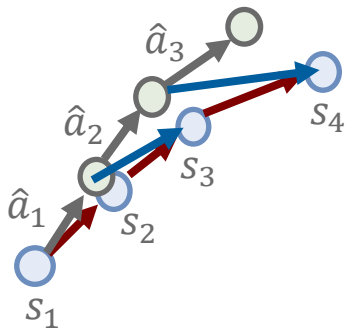
States visited by expert

States visited by the policy

Aggregating Corrective Behavior

Dagger: Dataset Aggregation

1. Roll out policy π_θ and collect states: $(s'_1, \hat{a}_1, \ldots, s'_t)$
2. Query/label expert action at visited states $\mathbf{a}^* \sim \pi_{\text{expert}}(\cdot \mid \mathbf{s}')$
3. **Aggregate** corrections with existing data $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}', \mathbf{a}^*)\}$
4. Update policy $\theta \leftarrow \arg\min_\theta \mathcal{L}(\pi_\theta, \mathcal{D})$



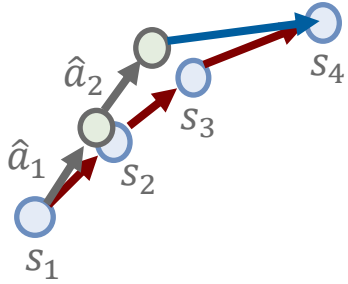
- ✅ Algorithm will converge $p_\pi(s) = p_{expert}$
- ✅ Achieves $O(\epsilon T)$ instead of BC $O(\epsilon T^2)$
- ❌ Hindsight labeling & expert querying difficult

Paradox BC works better if the data has more mistakes and recoveries!

Aggregating Online Interventions

Human Gated Dagger

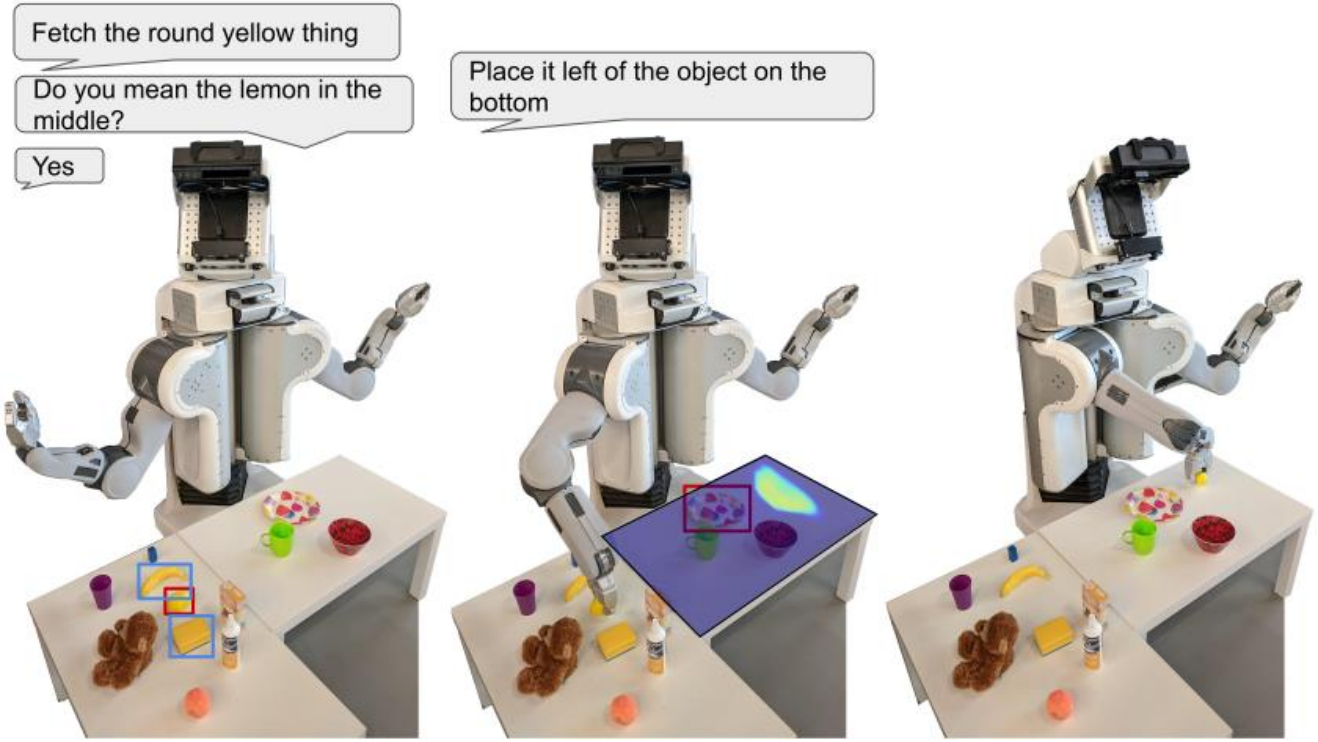
1. Roll out policy π_θ and collect states: $(s'_1, \hat{a}_1, \dots, s'_t)$
2. Expert intervenes at time t when policy makes mistake
3. Expert provides partial demo $(\mathbf{s}'_\mathbf{t}, \mathbf{a}^*_\mathbf{t}, \dots, \mathbf{s}'_\mathbf{T})$
4. **Aggregate** new demos $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}'_\mathbf{i}, \mathbf{a}^*_\mathbf{i})\}; i \geq t$
5. Update policy $\theta \leftarrow \arg\min_\theta \mathcal{L}(\pi_\theta, \mathcal{D})$



✅ Lets the human take control!

❌ How to detect when an intervention is needed?

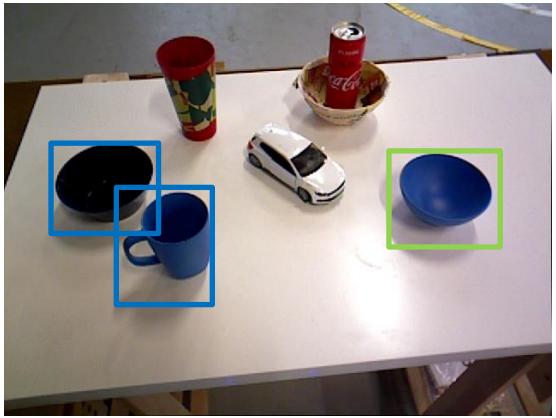
Case Study: Language Ambiguities



Composing Pick-and-Place Tasks By Grounding Language
Mees and Burgard, ISER 2020

Case Study: Language Ambiguities

"Pick up the blue stuff"



Do you mean this blue bowl on the table?

1. Train referential expression comprehension model with ranking loss

$$\mathcal{L}_{\text{rank}} = \sum_i \begin{bmatrix} \lambda_1 \max\big(0, m_1 + S(o_i \mid r_j) - S(o_i \mid r_i)\big) + \\ \lambda_2 \max\big(0, m_1 + S(o_k \mid r_i) - S(o_i \mid r_i)\big) \end{bmatrix}$$

max margin ranking loss

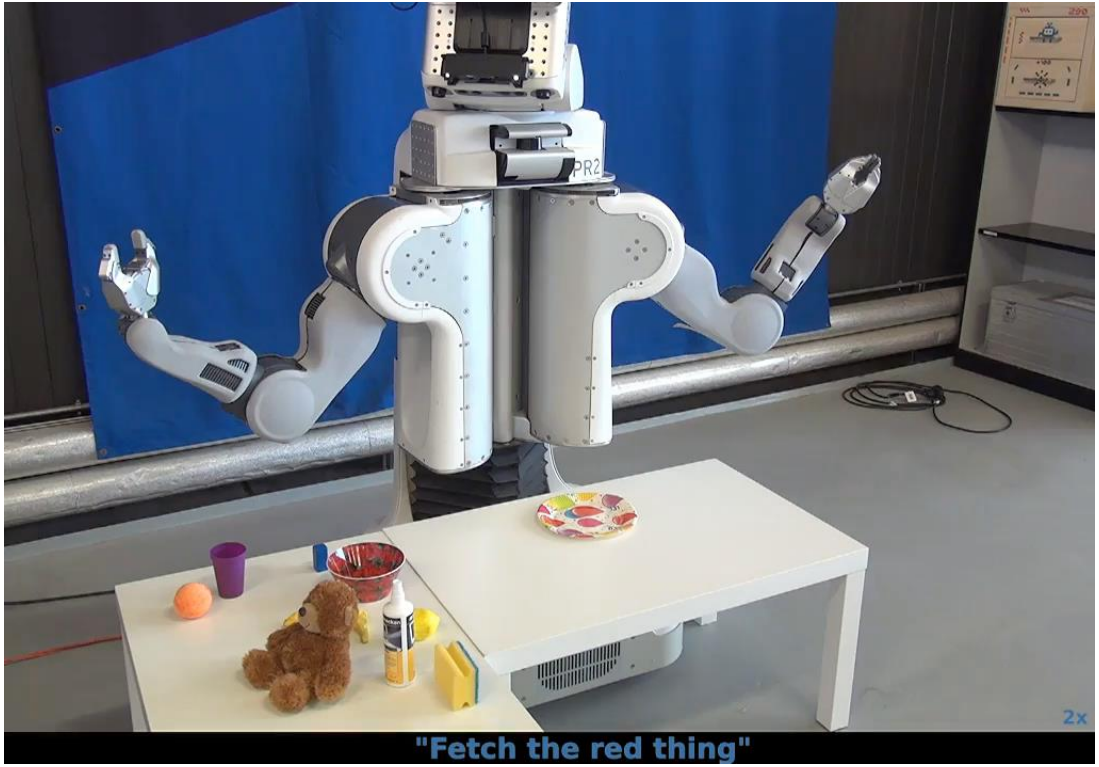
2. Train referential expression generation model

$$\mathcal{L}_{\text{gen}} = -\sum_i \log P(r_i \mid v_i)$$

$$\mathcal{L}_{\text{mmi}} = \sum_i [\lambda_3 \max(0, m_2 + \log P(r_i \mid v_k) - \log P(r_i \mid v_i))]$$

3. If at test time there are multiple objects within margin m, generate descriptions and ask user

Case Study: Language Ambiguities



Composing Pick-and-Place Tasks By Grounding Language
Mees and Burgard, ISER 2020

Why might we still fail to mimick the expert?

Non-Markovian behavior

$$\pi_\theta(a_t|o_t)$$

action depends only on current observation

$$\pi_\theta(a_t|o_1, \dots, o_t)$$

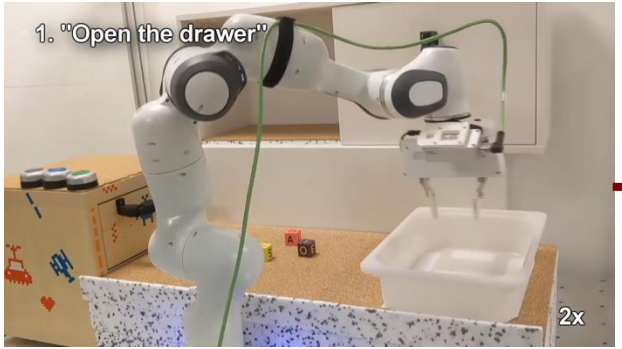
Human behavior might be affected by past observations, emotions, privileged information etc.

Case Study: Privileged observation

- Human observation ≠ robot's camera observations

Adding History to Policy

- Encode with some sequence to sequence architecture



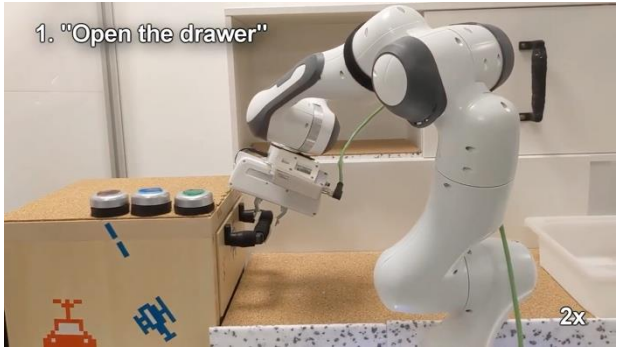
Seq2Seq Model

❌ Adding history to the policy **doesn't always make things better**!

=> causal confusion when doing IL with history

Causal Confusion

- Policy might infer **spurious correlations**!



1. Data: every time the robot opens the drawer, the gripper grasps the handle with 10 N
2. Model learns: if history shows a gripper force spike of 10 N, pull to open
3. Test time: gripper slips and only reads 2 N, the pull command is never triggered

❌ Policy thinks history of gripper sensor causes open drawer instead of the visual state!

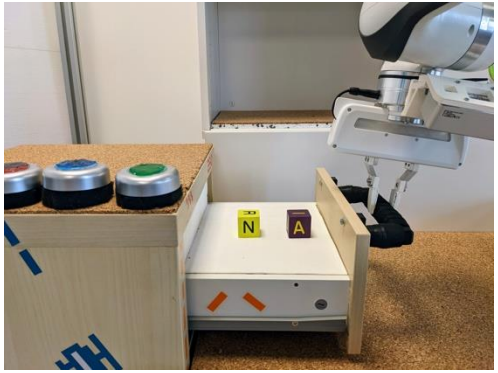
More analysis: “Causal Confusion in Imitation Learning” by de Haan et al., 2019.

Why might we still fail to mimick the expert?

Multimodal behavior

Stochasticity: Many (**infinite**) ways to close a drawer with a 7-DoF robot!

Expert Inconsistency: Humans operators might use different '**modes**' across trials

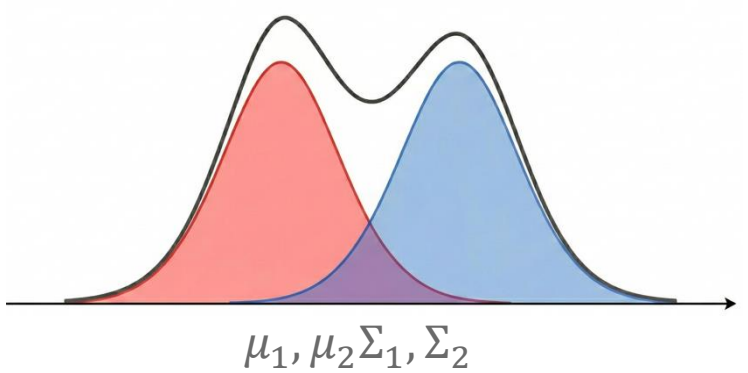


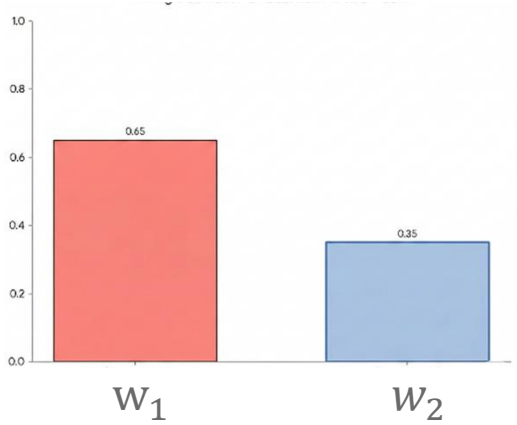
Multimodal Behavior

- Deterministic MSE BC policy will fail due to “**averaging**” of **modes**!

Mixture of Gaussian Distributions

- More expressive than a single gaussian
- Need to predefine number of gaussians





$$\pi(\mathbf{a} \mid \mathbf{o}) = \sum_i w_i \mathcal{N}(\mu_i, \Sigma_i)$$

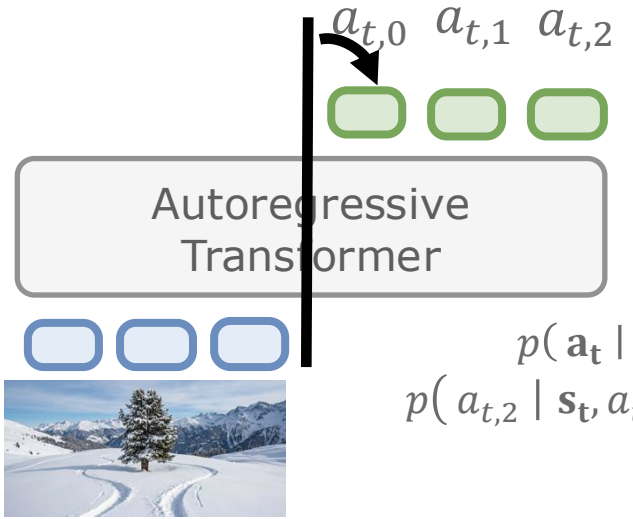
if you wanna do this for a humanoid with like 1000 modes this becomes quite complex. he mentioned something along the lines of "any distribution can be approximated by a mixture of gaussians" or so, which i would have to check

Autoregressive Discretization

- Discretization great for representing multimodal distributions action space as a grid, makes it a classification problem
- Impractical to scale to higher dimensions (exponential)
- Solution: **per-dimension discretization with sequence model**

linear in nr of dimensions, can use sequence model to do this

but sort of need LLM or VLA backbone to do this



$$p(\mathbf{a_t} \mid \mathbf{s_t}) = p(a_{t,0}, a_{t,1}, a_{t,2} \mid \boldsymbol{s_t}) = p(a_{t,2} \mid \mathbf{s_t}, a_{t,0}, a_{t,1})\, p(a_{t,1} \mid \mathbf{s_t}, a_{t,0})\, p(a_{t,0} \mid \mathbf{s_t})$$

Diffusion

- Models complex distributions over continuous variables
- We can replace images with robot actions, more on W6

fake dataset with images with varying level of noise, then train denoising network to predict cleaner version of images

Forward Process

$$\mathbf{x_{i+1}} = \mathbf{x_i} + \text{noise}$$

both per-dimension discretization and diffusion sort of use an iterative process, where the first is over action dimensions and the other over the denoising process

x_0 x_T

Generative Backward Process

learn $f(x_i) = x_{i-1}$ in practice $f(\mathbf{x_i}) = \text{noise}$, $\mathbf{x_{i-1}} = \mathbf{x_i} - f(\mathbf{x_i})$

Latent Variable Models

- Output still gaussian, but receives additional input
- Can represent “any” distribution
- Popular: conditional VAEs

Conditional Decoder

$$\pi(\mathbf{a} \mid \mathbf{o}, z) = f_\theta(\mathbf{o}, z)$$

$z_{\text{left}} \sim \mathcal{N}(0, \mathbf{I})$ $z_{\text{right}} \sim \mathcal{N}(0, \mathbf{I})$

Learning Multiple Tasks

Latent Variable Models are different from task conditioning

"one task"

$$\pi_\theta(a_t \mid s_t)$$

Single task
Behavior Cloning
(Pomerleau 1991)

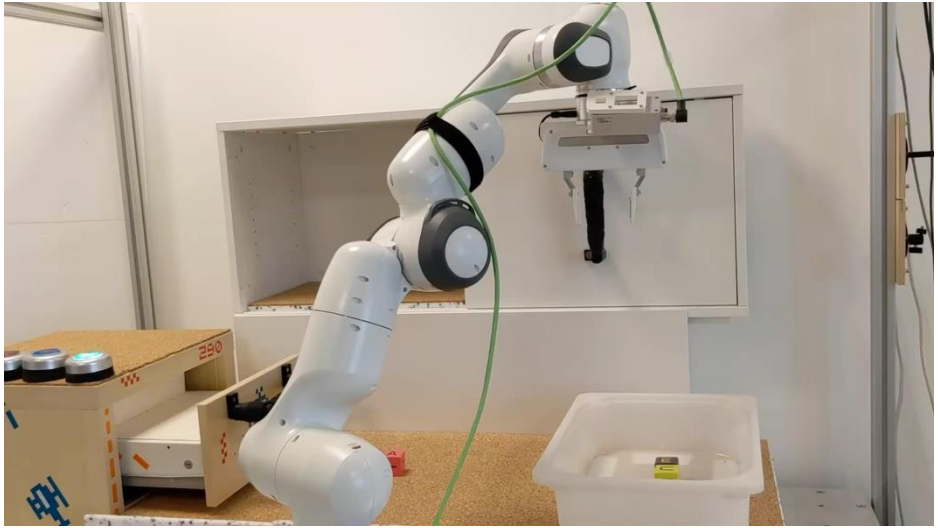
"multiple tasks"

$$\pi_\theta(a_t \mid s_t, \text{task id})$$

Task conditioned
Behavior Cloning
(Rahmatizadeh 2018)

Learning Multiple Tasks

"Move the sliding door to the left" $\pi_\theta(a_t \mid s_t, \text{task id})$



Did it succeed? Where is the threshold, 50%?

How to Scale Control to “Any” Tasks?

Tasks are often **continuous**, not discrete

“one task”

$$\pi_\theta(a_t|s_t)$$

Single task
Behavior Cloning
(Pomerleau 1991)

“multiple tasks”

$$\pi_\theta(a_t|s_t, \text{task id})$$

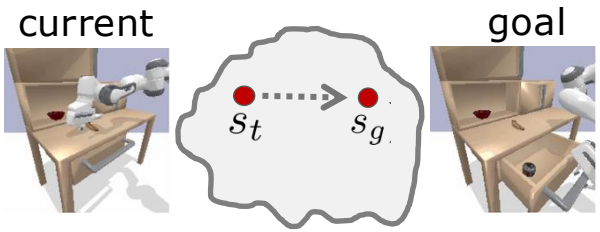
Task conditioned
Behavior Cloning
(Rahmatizadeh 2018)

“any tasks”

$$\pi_\theta(a_t|s_t, s_g)$$

condition on goal-reaching (initial and goal state)

Goal state conditioned
Behavior Cloning
(Lynch 2019)



Let’s put everything together!

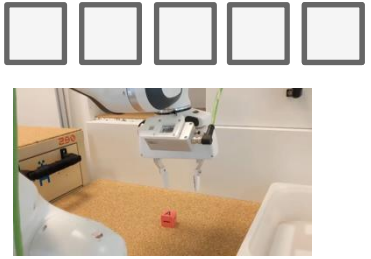
How can we learn any task with a latent variable model that handles multimodality?

Case Study: Play Data

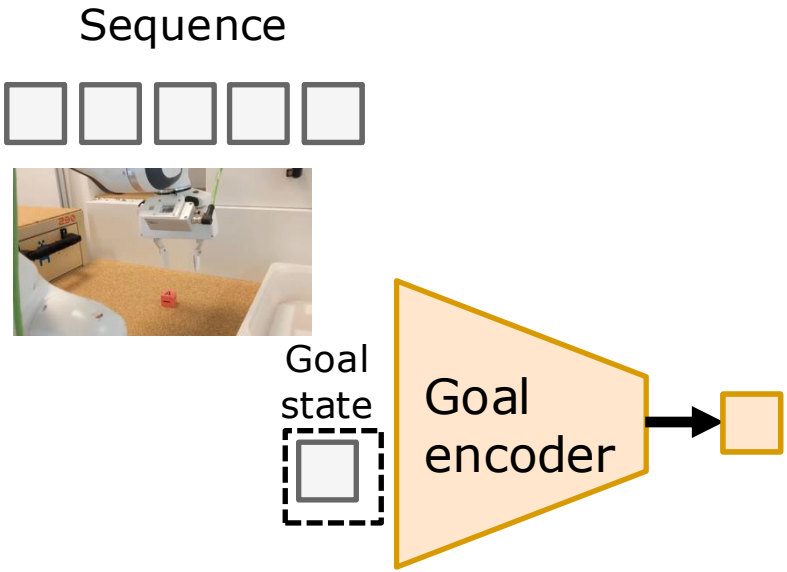
- No upfront tasks
- Scalable, reset-free data collection
- Rich and highly multimodal behaviors

Case Study: Learning Policies from Play Data

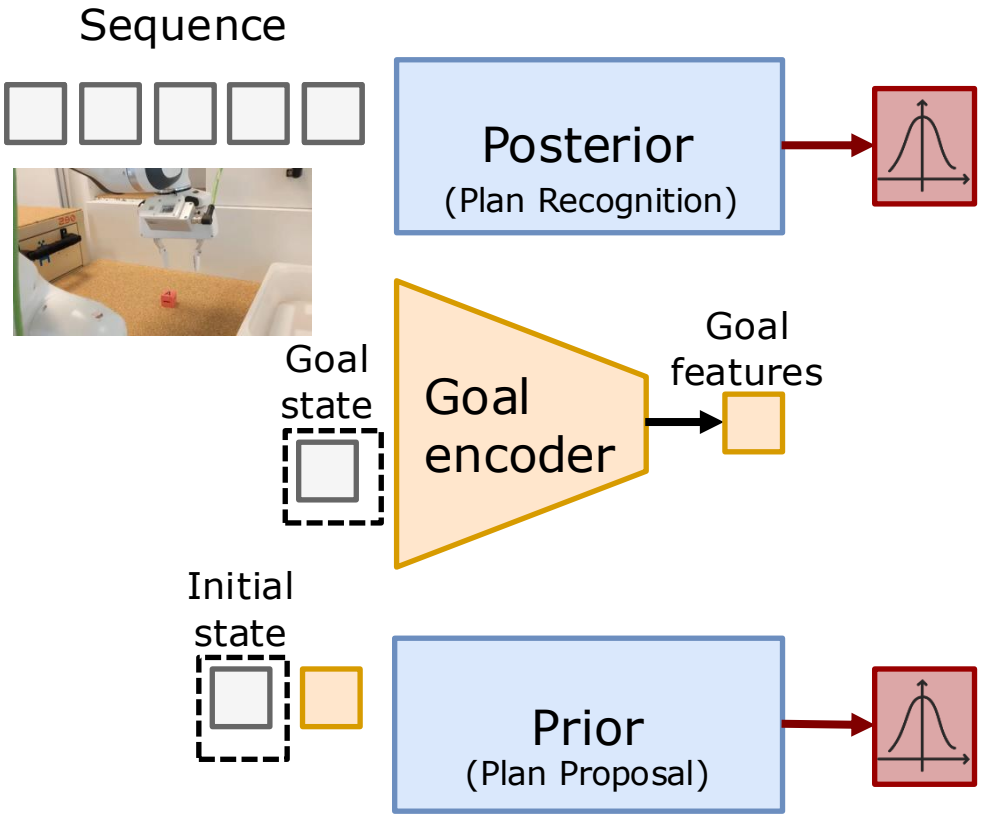
Sequence



Case Study: Learning Policies Play Data



Case Study: Learning Policies from Play Data



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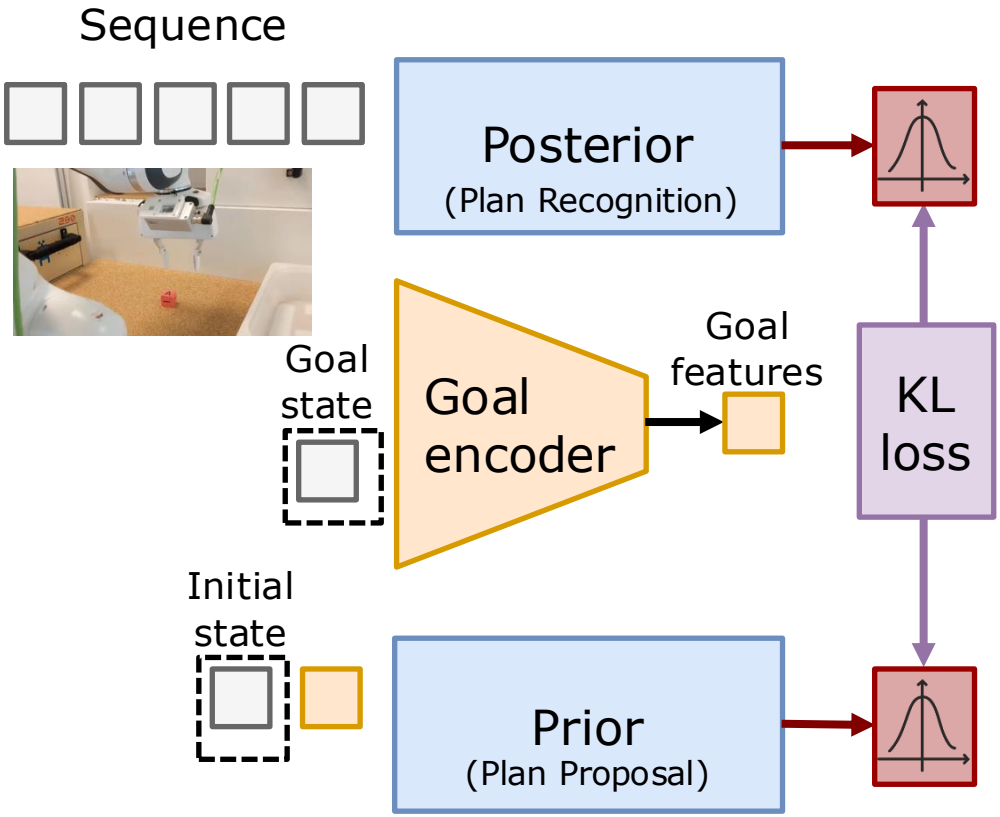

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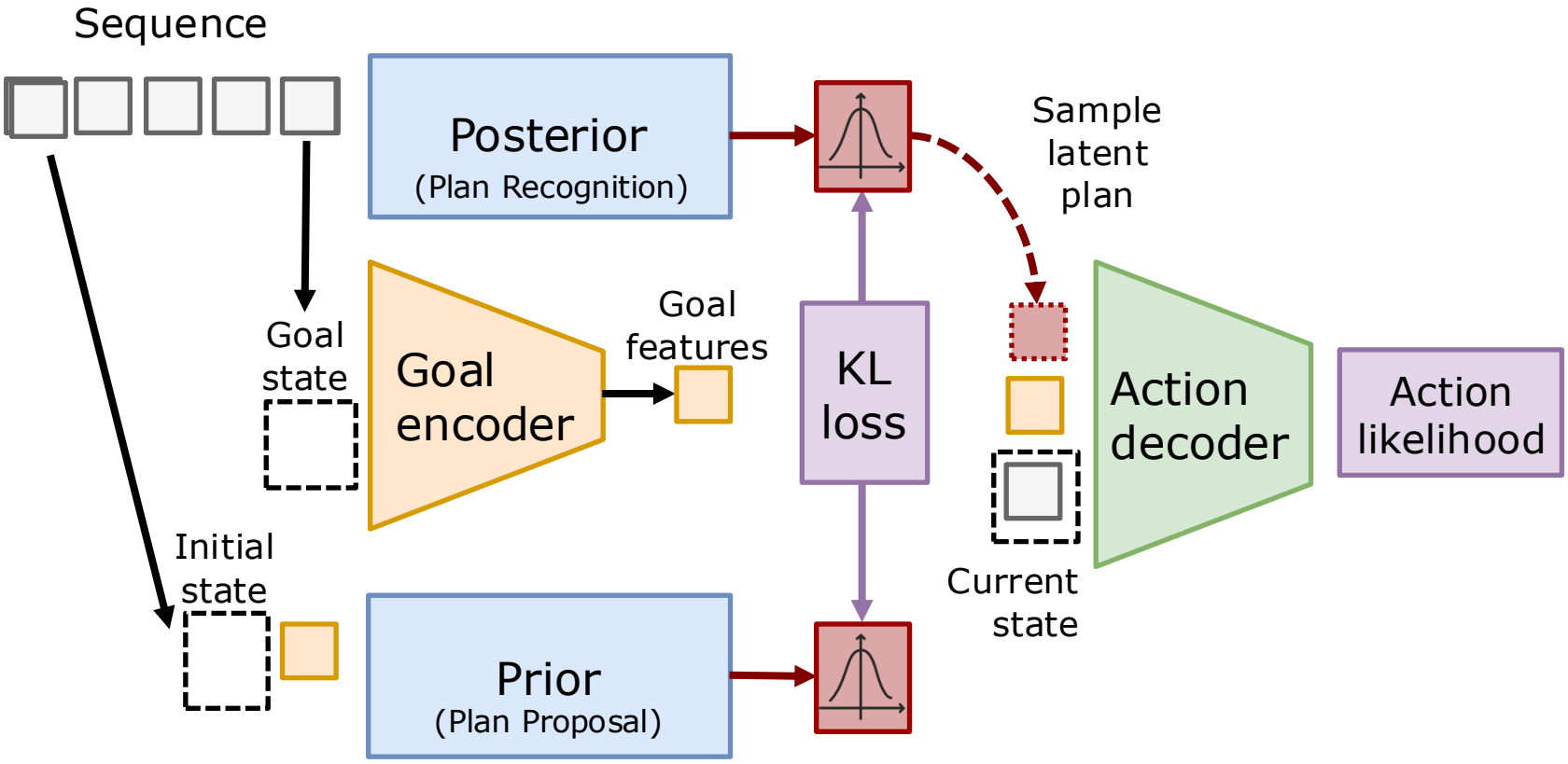
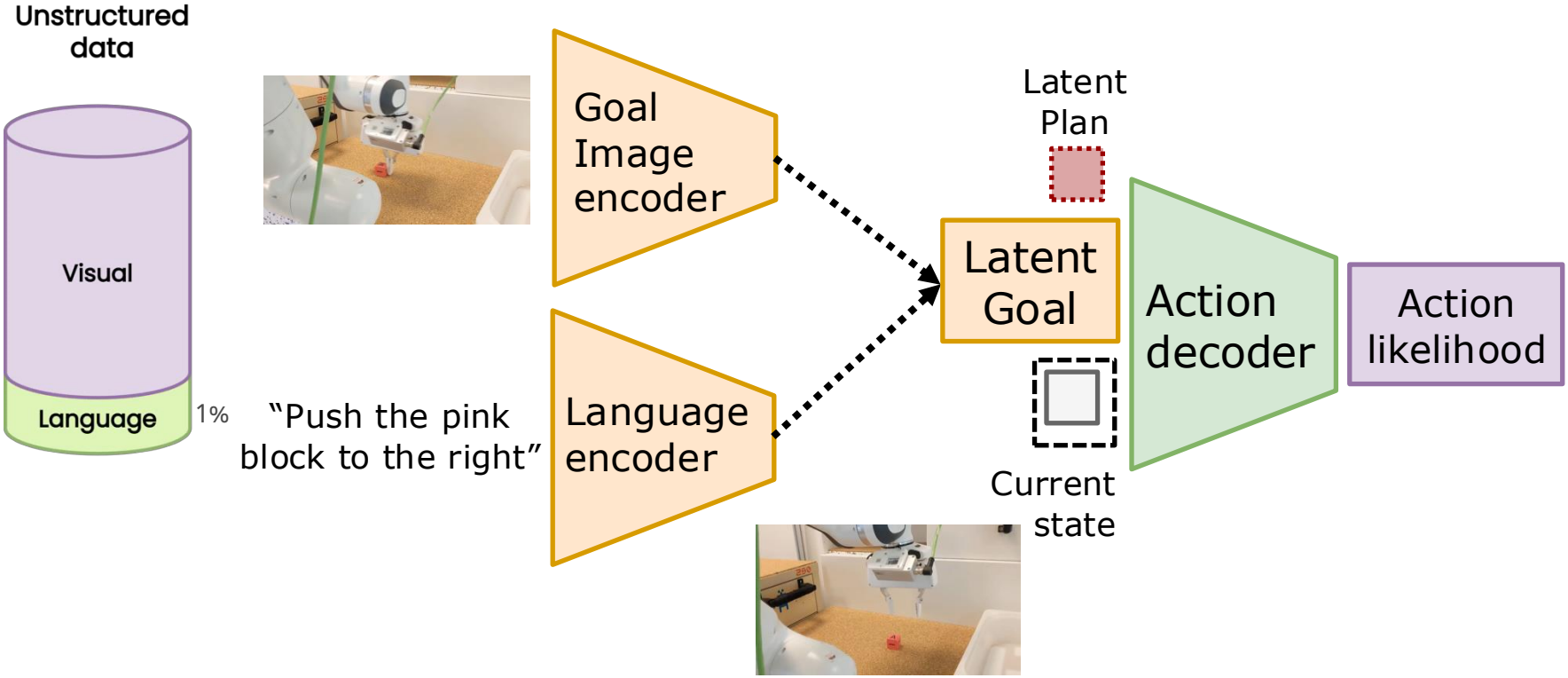


Case Study: Learning Policies from Play Data



The Math Behind Goal-Conditioned Latent Variable Models

trajectories; latent plans

$$p_\theta(\tau, z) \longrightarrow$$ when z continuous, marginalization becomes intractable

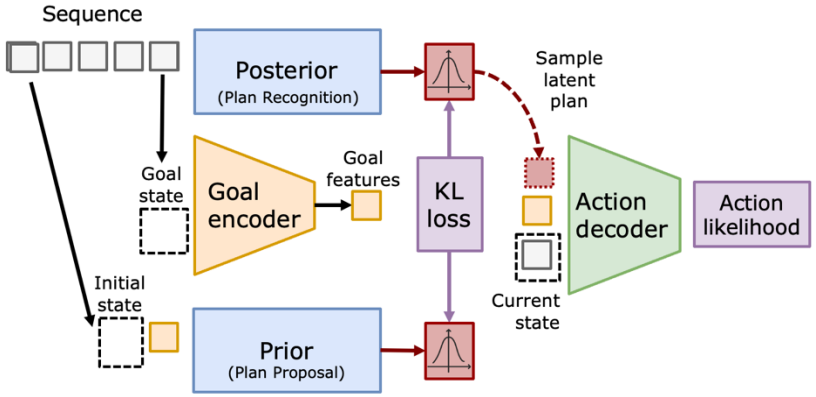
Optimize surrogate objective; variational lower bound of marginal log-likelihood

$$\log p_\theta(\tau) \geq -\underbrace{KL\left(q_\phi(z|\tau) \middle|\middle| p_\theta(z)\right)} + \underbrace{\mathbb{E}_{q_\phi(z|\tau)}\left[\log p_\theta(\tau|z)\right]}$$

Regularization:
are latent plan z close to our prior distribution over τ ?

Reconstruction:
how well does latent plan z explain the expert behavior τ ?

The Math Behind Goal-Conditioned Latent Variable Models



Posterior – Plan Recognition

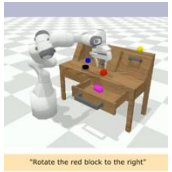
Prior – Plan Proposal

Action Decoder

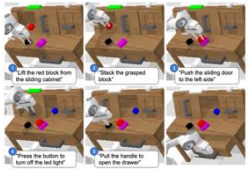
$$\log p_\theta(\tau|c) \geq -KL(q_\phi(z|\tau,c)\,||\,p_\theta(z|c)) + \mathbb{E}_{q_\phi(z|\tau,c)}\left[\log p_\theta(\tau|z,c)\right]$$

$$c \leftarrow (s_c, s_g)$$

My PhD: Natural Language + Learning from Play



CALVIN: A Benchmark for Language-conditioned Policy Learning for Long-horizon Robot Manipulation Tasks
Mees et. al., RA-L 2022
Best Paper Award RA-L 2022



What Matters in Language Conditioned Imitation Learning over Unstructured Data
Mees, et al. RA-L 2022.

Grounding Language with Visual Affordances over Unstructured Data
Mees, et al. ICRA 2023.
Finalist Best Paper Award ICRA 2023

Affordance Learning from Play for Sample-Efficient Policy Learning
Borja*, **Mees*** et. al., ICRA 2022

Latent Plans for Task Agnostic Offline Reinforcement Learning
Rosete*, **Mees*** et. al., CoRL 2022

Conclusion: Behavioral Cloning Failures

Distribution Shift

- Dagger
- Human Gated Dagger

↓

Detect when to intervene

Non-Markovian Behavior

- Privileged Observation
- History & Causal Confusion

Multimodal Behavior

- Mixture of Gaussians
- Autoregressive Discretization
- Diffusion
- Latent Variable Models

Thank you for your attention

References

- Uni Freiburg, Deep Learning Lab
- UC Berkeley Deep RL
- Stanford University, Deep RL
- Cornell Robot Learning